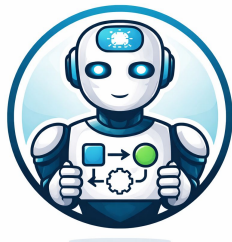


Agentic AI for Tech Professionals

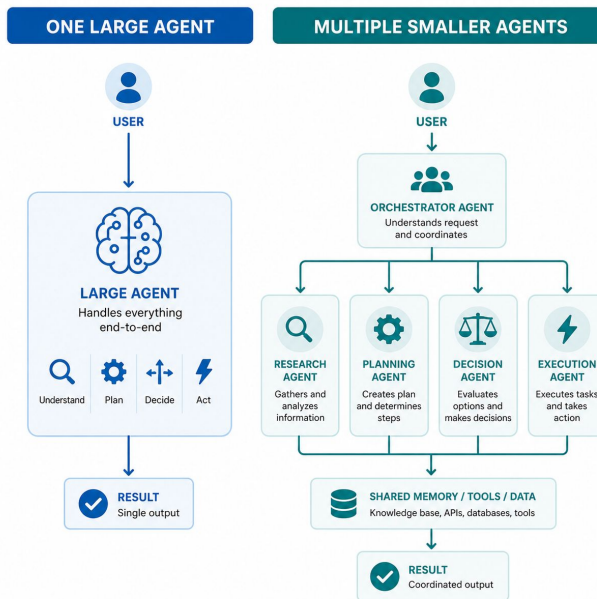
Multi-Agent Architecture



From One Agent → Many Agents

Break complex systems into smaller specialized agents

- One agent = limited capability
- Multiple agents = specialized roles
- Each agent solves one problem
- Easier to debug and scale
- Mirrors real-world teams



From One Agent → Many Agents (2)

A single AI agent trying to do everything becomes:

- messy
- unpredictable
- hard to maintain

Instead, we split into **specialized agents**.

Think of a company:

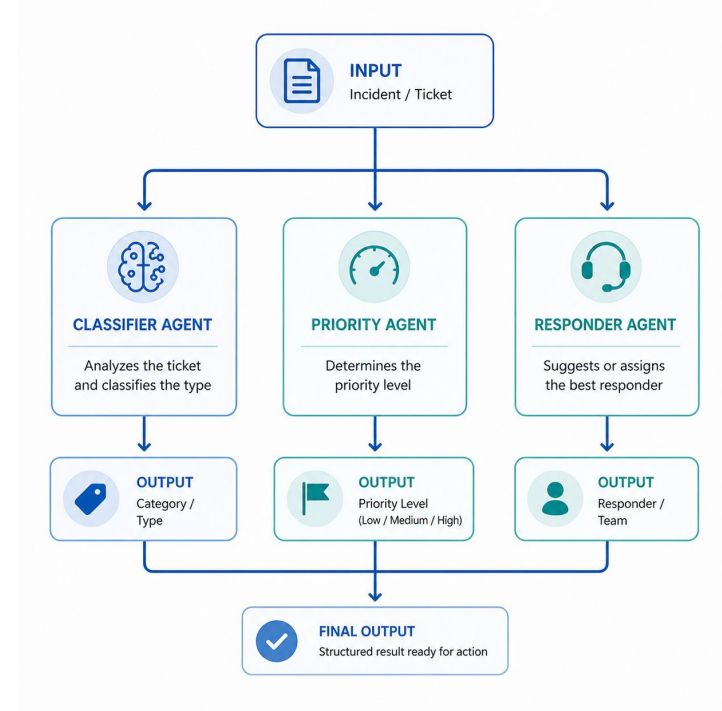
- One person does everything → chaos
- Multiple roles → efficient

Same for AI systems.

Specialized Agent Roles

Each agent has a clear responsibility

- Classifier Agent → identifies issue type
- Priority Agent → assigns urgency
- Responder Agent → generates response
- Each agent has its own prompt
- Outputs are structured (JSON)



Specialized Agent Roles

We don't want one AI doing everything.

Instead:

Input:

“Machine overheating”

Classifier Agent:

👉 Equipment

Priority Agent:

👉 High

Responder Agent:

👉 “Stop machine and check cooling system”

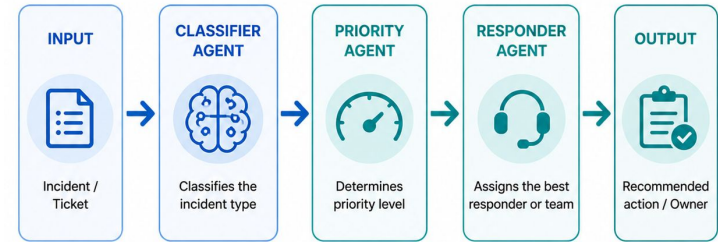
Each agent is:

- focused
- predictable
- easier to improve

Agent Collaboration Flow

Agents work together in sequence

- Output of one agent = input to next
- Forms a pipeline
- Enables step-by-step decision
- Reduces AI confusion
- Improves accuracy



Agent Collaboration Flow (2)

Example:

Input:

“Repeated login failure across multiple stations”

System:

- Classifier → IT
- Priority → High
- Responder → escalate to IT team

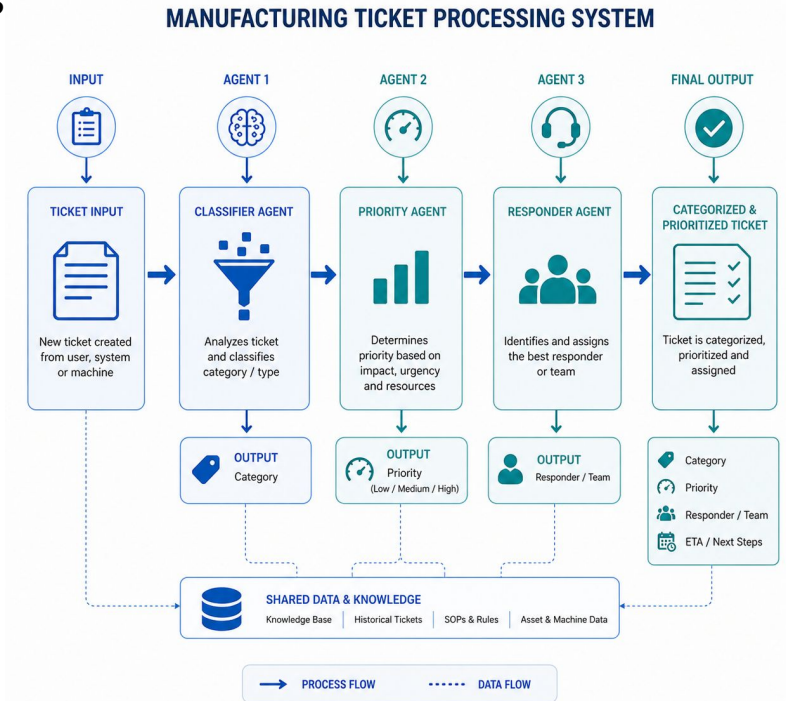
This is no longer a chatbot.

👉 This is an **AI workflow system**

Use Case Mapping — Multi-Agent System

Apply multi-agent design to manufacturing incidents

- Input: incident ticket
- Classifier → issue category
- Priority → urgency level
- Responder → action message
- Output: structured + actionable



Activity 1 : Multi-Agent System

“Repeated login failure across production terminals”

Which agent should:

- identify issue type?
- assign urgency?
- generate operational response?

Activity 2: Design your own Multi-agent workflow

“Repeated inventory mismatch across warehouse systems”

Task:

Design:

- at least 3 agents
- each agent's responsibility
- workflow sequence

Activity 3 : Build your own agent

Scenario Options

- Production machine overheating
- Supplier invoice duplication
- Repeated authentication failure
- Data quality issue
- Customer complaint routing

Task:

Map the workflow using:

- 👉 Input
- 👉 Classifier Agent
- 👉 Priority Agent
- 👉 Response Agent
- 👉 Final Output